

SEQUENTIAL SYSTEMS

Phase VII: Advanced Architectures

(Ring Counters & Sequence Generators)

Comprehensive Academic Master Addendum

Topics Covered (Based on STC Sir's Notes):

Programmable Up/Down Ripple Counters

Shift Register Counters

The Ring Counter (Mod-N)

The Johnson Counter (Mod-2N)

Arbitrary Sequence Generators

Methodology: Pure $\text{\LaTeX}$ Compilation

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Chapter 1

Programmable Up/Down Counters

1.1 The Dual-Direction Problem

In Phase III, we constructed an asynchronous ripple up-counter using negative-edge triggered T-Flip-Flops. We established a fundamental rule of ripple architecture:

- If the clock is driven by the Q output of the previous stage, the circuit counts **UP**.
- If the clock is driven by the $\overline{Q}$ output of the previous stage, the circuit counts **DOWN**.

However, many engineering applications require a counter that can dynamically reverse direction on command. To achieve this, we introduce a **Mode Control Input (M)**.

1.2 The Mode Control Logic Matrix

We must design a combinational logic circuit between each flip-flop that acts as a directional switch. Let M be the control line. Let Y be the signal routed to the clock pin of the next stage.

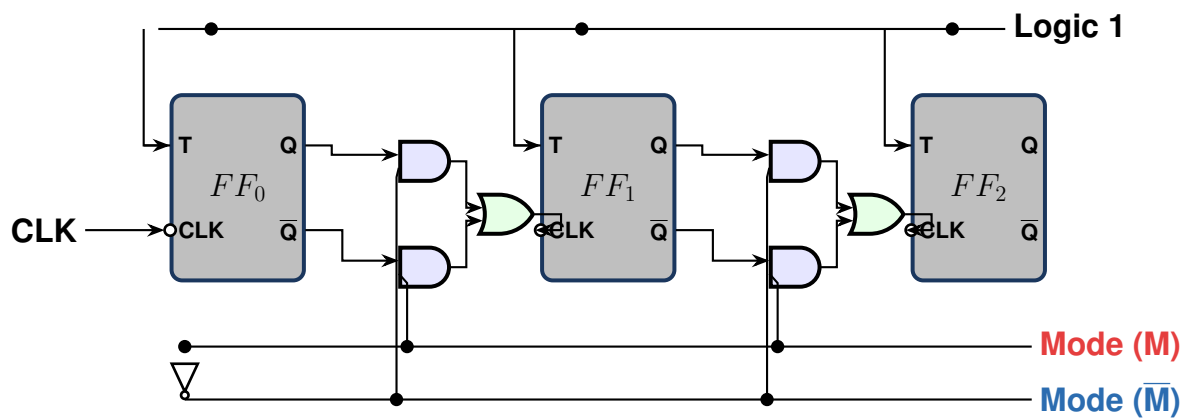
- When $M = 0$: We want $Y = Q$ (Up Counting).
- When $M = 1$: We want $Y = \overline{Q}$ (Down Counting).

We can express this logic mathematically using a 2x1 Multiplexer equivalent circuit (an AND-OR-INVERT topology):

$$Y = (\overline{M} \cdot Q) + (M \cdot \overline{Q}) \quad (1.1)$$

1.3 Architectural Implementation

The circuit below demonstrates a 3-Bit Asynchronous Up/Down Counter. The Mode line M acts as a universal bus. By toggling M between 0 and 1, the logic gates physically reroute either Q or $\overline{Q}$ into the subsequent clock pins.



Chapter 2

Shift Register Counters

2.1 Beyond the Binary Count

Traditional counters utilize T or JK flip-flops to progress through dense binary sequences (000, 001, 010). However, some specialized applications require sparse, circulating sequences.

By taking a standard **Shift Register** (composed of D-Flip-Flops) and routing the output of the final stage directly back into the input of the first stage, we create a closed-loop system known as a **Shift Register Counter**.

2.2 The Ring Counter (Mod-N)

The simplest variation is the **Ring Counter**. The Q output of the last flip-flop is wired directly to the D input of the first flip-flop.

2.2.1 Initialization & Operation

A Ring Counter cannot start from a state of all zeroes. If the register holds 0000, shifting it will simply circulate the zeroes endlessly. To function, the circuit requires an initialization pulse (using the asynchronous PRESET and CLEAR pins) to forcefully inject a single '1' into the array. For example, initializing a 4-bit array to **1000**.

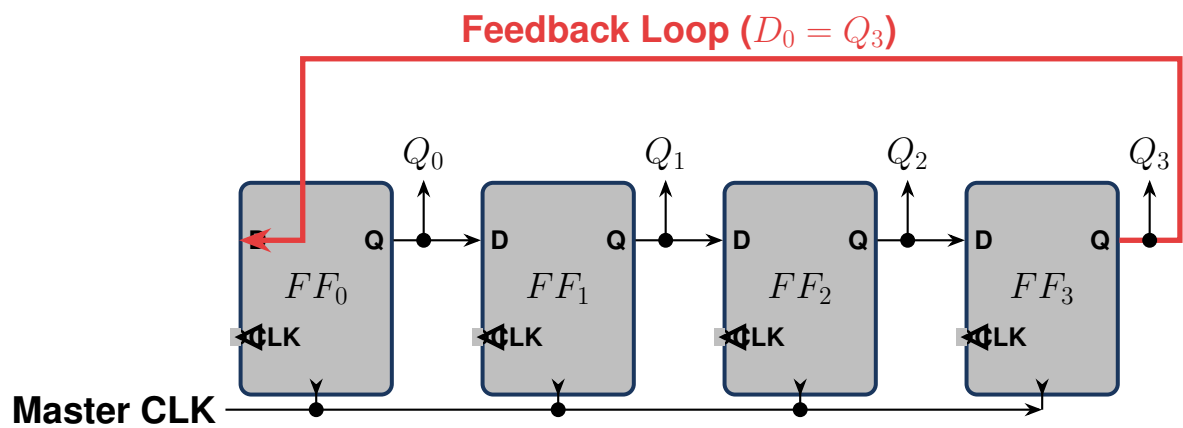
Upon receiving clock pulses, this single '1' behaves like a baton being passed down a relay race.

- Clock 0: 1000
- Clock 1: 0100
- Clock 2: 0010
- Clock 3: 0001
- Clock 4: 1000 (The '1' loops back from Q_3 to D_0)

Because it requires N clock pulses to complete one full circulation of the '1', an N-bit Ring Counter is mathematically classified as a **Modulo-N Counter**.

2.2.2 Ring Counter Schematic (4-Bit)

Notice how the feedback wire loops the Q_3 output completely around the array and directly into the D_0 input.



2.3 The Johnson Counter (Twisted Ring)

The primary inefficiency of the standard Ring Counter is that it requires 4 flip-flops to achieve only 4 unique states. It is a massive waste of silicon logic potential.

To extract more utility from the hardware, engineers invented the **Johnson Counter** (also known as the Twisted Ring or Switch-Tail Counter). Instead of feeding Q_{MSB} back to D_0 , we twist the feedback wire and route the **inverted output** ($\overline{Q}_{MSB}$) into D_0 .

2.3.1 Initialization & Operation

The Johnson Counter initializes with a full flush of zeroes: **0000**. Because $Q_3 = 0$, the inverted output $\overline{Q}_3 = 1$. The feedback wire routes this '1' into D_0 . As the clock pulses, '1's begin to fill the register from the left. Once the register is completely full of '1's, the logic reverses, and '0's begin to fill the register.

Modulus Amplification

By simply inverting the feedback wire, the exact same 4 flip-flops now generate **8 distinct states** instead of 4. An N-bit Johnson Counter is mathematically classified as a **Modulo-2N Counter**.

Pulse	Q_0	Q_1	Q_2	Q_3	
0	0	0	0	0	
1	1	0	0	0	
2	1	1	0	0	
3	1	1	1	0	
4	1	1	1	1	
5	0	1	1	1	
6	0	0	1	1	
7	0	0	0	1	
8	0	0	0	0	

Chapter 3

Sequence Generators

3.1 The Philosophy of Arbitrary Sequences

Thus far, we have synthesized machines that count sequentially (0,1,2,3), and machines that shift data linearly.

However, advanced communications protocols (like radar telemetry or digital cryptography) frequently require hardware to generate an *arbitrary, non-sequential bitstream* repeatedly. For example, a system may require a dedicated microchip to continuously broadcast the binary sequence: **1, 0, 1, 1, 0, 1...**

This specific architecture is called a **Sequence Generator**.

3.2 Synthesis Methodology

A Sequence Generator can be synthesized using standard FSM design principles (the 6-Step STC methodology) or by heavily modifying a Shift Register.

If we desire to generate a sequence N bits long, we configure a synchronous

array of flip-flops. We construct a combinational logic matrix that reads the present state of the array and calculates the exact next bit required in the sequence. That calculated bit is forcefully routed into the input of the first flip-flop.

The Role of Multiplexers in Generation

In advanced implementations (as noted in STC Sir's advanced lectures), instead of using K-Maps to derive complex AND/OR gate logic, engineers simply use an **8x1 or 16x1 Multiplexer**. The outputs of the counter are wired directly to the MUX select lines. The programmer hardwires the MUX input pins to V_{cc} (1) or GND (0) matching the desired sequence. The MUX acts as a purely hardware-based "Look-Up Table" (LUT), streaming out the exact sequence at maximum clock speed.

***** END OF PHASE VII ADDENDUM**
